# Nicholas Henry

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## EDUCATION

**Northeastern University** Boston, MA
*Bachelor of Science in Mechanical Engineering and Computer Science* *May 2029*
**Coursework**: Diff Eq and Lin Alg, Discrete Structures, Program Design and Implementation, Statics *GPA: **3.79***
**Activities**: Northeastern Electric Racing (FSAE), Northeastern Satellite Lab, Dean's List
Semester Abroad at Queen's University Belfast in Northern Ireland

## SKILLS

**Languages:** Python, Java, C#, JavaScript, SQL, HTML/CSS, TypeScript, C++
**Frameworks & Libraries:** React, Next.js, Node.js, Tensorflow, NumPy, Flask, Dash, Tailwind, PostgreSQL
**Tools & Platforms:** Git, Docker, Github Actions, Redis, Supabase, YoloV8, Cursor, Linux, ROS, CI/CD, Claude Code, Raspberry Pi, SolidWorks, MATLAB
**Certifications & Training:** IBM Artificial Intelligence Fundamentals, Microsoft C# Certification, GitHub Foundations

## EXPERIENCE

**Citizens Bank** - *Python, UiPath, Git* Westwood, MA
*Incoming Automation Developer Co-op* Jan 2027 – Jun 2027
- Joining the automation engineering team to develop scripts and tooling that streamlines internal banking operations

**Northeastern Electric Racing** - *Python, Dash, React, Docker, Git, Redis, Grafana* Boston, MA
*Lead Software Engineer for Performance Analysis Team* Jan 2026 – Present
- Develop a parametric vehicle simulation platform (Python backend and React/Dash frontend) used by 120+ engineers across 8 subteams to evaluate how geometry and setup changes effect lap times on custom built tracks
- Pioneered the quasi-steady-state evaluator that calculates maximum cornering speed per track segment, writing results to Redis with a 24 hour TTL so users can visit and compare recent runs without re-simulating
- Lead 15 developers through weekly meetings, review weekly PRs, and validation using Grafana telemetry to 3% error

**Students for the Exploration and Development of Space** - *Python, ROS, Git, Linux* Boston, MA
*Software Engineer for Network Exploration Swarm Toolkit Team* Jan 2026 – Sep 2026
- Program for a team to create a swarm of C.O.B.R.A (Crater Observing Bio-inspired Rolling Articulator) snake robots to explore and develop multi-agent collaboration in order to complete teamwork based exploration tasks
- Created the camera sensor ecosystem consisting of a simultaneous mapping program for multi-agent systems

**SEONeeded** - *Wordpress, SEOMoz, Semrush* Miami, FL
*Web Optimization Engineer* Oct 2025 – Aug 2026
- Performed client-facing tasks in order to drive more traffic increasing click-on rate and sales by 17% for 15 clients
- Adapted to the new wave of generative engine optimization space by learning what makes LLM models reference client's websites and how certain worded prompts get tailored responses, recommendations, and links to sources

*Search Engine Optimization Intern* Jun 2025 – Sep 2025
- Drove a 54% increase in website traffic through targeted optimization by leveraging WordPress and keyword research

## PROJECTS

**SheepFinder -** *TypeScript, React, Next.js, PostgreSQL, Tailwind* May 2026 – Present
- Launched a lost animal reporting app based off studying abroad in Northern Ireland, designing the structured marking system (type, color, and location) based on how farmers identify sheep markings between farms and towns
- Designed a weighted matching engine in TypeScript that scores sightings against registered animals by species, color, and markings and ranking into strong, likely, and possible confidence tiers before notifying users
- Reunited over 10 lost animals with their owners after designing the app around a focus group of farm owners

**Bro Look At My Wheat (HackHarvard Project) -** *Python, Factory, Photon, Arduino, YoloV8, Cursor* Oct 2026
- Trained a drone to autonomously search for wheat with a YoloV8 model, determine its close, then return to an april tag
- Built a data pipeline that took the images and weather data gathered and matched them to satellite info in the area and analyzed it using chalk data agents to then send messages to the farmer via sms of what their next steps should be
- Run on virtual environments through Factory so there's no physical overhead for farmers and faster data processing

**Text Transcriber -** *Python, TensorFlow, Google APIs, Replit* Jul 2025
- Trained a CNN with TensorFlow on 100,000+ handwritten character images and obtained 96% validation accuracy
- Tested writing and signage with a group of visually impaired participants to reduce caretaker dependency; results from the limited cohort were an 88% rate of understanding the written text which greatly improved independence